ABHINAY RAJ

Senior Frontend Engineer | React, TypeScript, Electron

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Frontend engineer with 4+ years of software development experience building complex, real-time meeting and webinar applications with React, TypeScript, and Electron. Focuses on frontend architecture, rendering performance, scalability, and browser-level problem solving. Takes features from prototype to production and customer rollout through hands-on ownership and cross-team collaboration.

3–4 sec Join time, down from ~15 sec	**3,000+** Webinar attendees · eliminated ~10s stalls	**32+ packages** UI migration · 60–70% less manual effort	**AI agents** PR review · tests · Jira-to-code · issue analysis

TECHNICAL SKILLS

Frontend Engineering: JavaScript, TypeScript, React, HTML, CSS, Redux, Electron, component architecture, responsive UI

Performance & Browser Engineering: Rendering optimization, code splitting, lazy loading, event batching, virtualization, Web APIs, AudioWorklet, Babel

Real-Time & Media: WebRTC, WebSocket, WebTransport, Web Audio API, HLS, LL-HLS, CEF

Testing & Engineering Quality: Unit and integration testing, end-to-end UI automation, production debugging, logging, diagnostics

Backend & Developer Tools: Node.js, Express, REST APIs, Rust, AWS, Docker, Git, continuous integration and delivery (CI/CD), AI engineering workflows

EXPERIENCE

Frontend engineering → performance at scale → technical leadership

GoTo Technologies (Remote)

Software Development Engineer — May 2024 – Present

- Reduced application join time from ~15 seconds to **3–4 seconds** by loading only required code and prefetching shared resources and feature data before users clicked Join.
- Eliminated ~10-second people-panel stalls for **3,000+ webinar attendees** by reducing real-time communication (RTC) signaling, batching UI updates, and moving event calculations off the main thread. Rendered attendee, chat, and Q&A lists on demand to keep Q&A responsive with **1,000+ questions**.
- Cut manual migration effort by **60–70%** while coordinating a 1.5-year Chakra UI-to-Chameleon migration across **32+ packages** with 5 engineers. Built a Claude-assisted workflow with codebase context and reusable skills to update packages in parallel.
- Built team-adopted AI agents for pull-request review, test generation, and Jira-to-code implementation. The review agent complements manual checks with a broad pass across changes, helping surface issues that might otherwise be missed; added continuous-integration checks for affected packages.
- Built Attendee Attention and its post-session reports for organizers; contributed to Focus Speaker and Q&A.
- Coordinated an 8-person team and collaborated with RTC engineers to combine organizer feeds through Chromium Embedded Framework (CEF) into one LL-HLS webcast, synchronized with audio/video, chat, Q&A, and live actions.
- Took Unmute Assistant from hackathon prototype to customer rollout, using AudioWorklet to move audio processing off the main thread. Worked with 4–5 teammates over 3–4 months to fix webinar audio failures, including inaudible streams on low-end machines, by starting audio streams one at a time, checking playback, handling permissions and errors, retrying failed connections, and improving diagnostics.
- Built Electron tab capture and pop-out windows. Used DOM reparenting to keep responsive meeting controls mounted across layout changes without remounting React components, preserving their state and avoiding unnecessary lifecycle work. Enabled data sharing between pop-outs and the main app, and used Babel build analysis to load only needed styles.
- Built responsive meeting and webinar interfaces from Figma designs using the Figma MCP server; documented design options and proof-of-concept findings with engineers and architects to support consistent implementation and handoffs.

Associate Software Development Engineer — Aug 2022 – May 2024

- Built Quick Polls and Q&A for meetings and webinars. Added attendee questions, organizer answers, panelist assignment, and public or private replies; also contributed to breakout rooms and webinar setup.

- Built synchronized YouTube, MP4, and PowerPoint sharing with playback and slide controls, a laser pointer, and drawing tools; improved YouTube sync despite limited player controls.
- **Hackathon prototypes and customer rollout:** Prototyped PPT sharing, shared-screen drawing, remote control, GoTo Assistant, and AI meeting summaries for late joiners and post-meeting host emails. Took chat tagging through webinar rollout with reactions and pin/unpin; PowerPoint sharing and drawing also reached customers.

Software Development Engineer Intern **Feb 2022 – Aug 2022**

- Created end-to-end UI regression tests for GoTo Webinar on web and Electron.
- Progressed from QA and automation—investigating defects and learning the product—to feature development during the six-month internship.

Co-Founder / Ex-CTO — **ONI Stays** **2020 – 2021**

- As CTO and sole engineer, took a co-living product from concept to launch, designing its architecture and building tenant, partner, and admin tools.
- Delivered REST APIs, integrations, and backend services; set up monitoring, CI/CD, and AWS/Docker deployment.

SELECTED PROJECTS

- **Real-Time Collaborative Canvas:** Built a React/TypeScript canvas with room state and live presence. Its Rust/Tokio WebTransport server uses reliable streams for updates and QUIC datagrams for cursor movement.
- **Streamify — Interactive Live Streaming:** Combined CEF media mixing, MediaMTX LL-HLS delivery, and synchronized interactive slides. Added translated screen-share content with selectable text and links.
- **React WebRTC Multi-Camera & Screen Sharing:** Built a React and TypeScript app for multiple users to share webcams and screens simultaneously, using Socket.IO signaling and RTCDataChannel messaging.

EDUCATION

Manipal Institute of Technology, Manipal **2022**
B.Tech, Computer Science